



function :

Easy to read

Easy to modify

Avoids rewriting of same code

Easy to debug

Better memory utilization

function in a program is to **save memory space** which becomes appreciable when a function is likely to be called **many times**.

However every time a function is called, it takes **lot of extra time** in executing a series of instructions for tasks such as **jumping** to the functions, **saving registers**, **pushing arguments** into the stack and **returning** to the **calling function**.

So when function is small it is **worthless** to spend so much **extra time** in such tasks in **cost of saving comparatively small space**.

To eliminate

To eliminate the cost of calls to small functions, C++ proposes a new feature called **inline function**.

An **inline function** is a function that is expanded in line when it is invoked. It means that -

The compiler replaces the function call with the corresponding function code.



inline function :

syntax : write inline before function declaration or definition

Important :

- inline is a request not a command.
- The benefit of speed of inline functions reduces as the functions grows in size.
- So the compiler may ignore the request in some situation.

few of them :

- function containing loops, switch, goto.
- function with recursion.
- containing static variable.

Program: `#include <iostream>`

`inline void fun();`

`void main()`

`{`

`cout << "You are in main";`

`fun();`

`}`

→ inline replace function call with function code

`void fun()`

`{`

`cout << "You are in fun";`

`}`